You have received this email from jasper@uci.edu in response to your Registration.

Registration Receipt

We’re looking forward to seeing you at HydroML 2025!

Registration:	**729090**
Store:	**HydroML**
Date/Time:	**May 5, 2025 at 11:42:10 AM PDT**
Total:	**$850.00**

Billed To:
Shubhendu Kumar Singh
4205 Alabama Ave
Apt B
Los Alamos, NM
87544
United States

Contact Email:
shubhsingh@lanl.gov

Payment Information:

Payment Type:	Credit Card
Credit Card Number:	xxxxxxxxxxx1004
Reference Number:	20250505000175
Card Type:	American Express

*** Card Not Present ***

Shipping Information:

Shipping Information:	704182
Delivery Method:	None

Item	Unit Price	Detail Total
General Registration	$500.00	$850.00
Lodging Option Add-On: Shared Deluxe Chalet	$350.00	
Name: Shubhendu Kumar Singh		
Institution: Los Alamos National Laboratory		
Phone: +1 716 533 0952		
Email: shubhsingh@lanl.gov		
Do you have any dietary restrictions? The Lodge will do its best to accommodate. : No restrictions		
Emergency Contact Name: Jon Schwenk		
Emergency Contact Phone Number: +1 505 717 5103		
	Subtotal:	**$850.00**
	Tax:	**$0.00**
	Total:	**$850.00**

Return Policy Agreement:
If you need to cancel your participation, we will refund 95% of your registration fee through Tuesday, May 13.

For questions, comments, or Registration status, send email to jasper@uci.edu and refer to Registration 729090. Visit us online.

Dates: May 27 – May 29, 2025 (all times PDT)

Location: Lake Arrowhead Lodge

Address: 850 Willow Creek Road, Lake Arrowhead, CA, USA, 92352

Host: Dr. Jasper A. Vrugt, UC Irvine, Co-host: Dr. Efi Foufoula-Georgiou, UC Irvine

Talks are in Pineview and Posters are in Lakeview

Tuesday, May 27, 2025

4.00 – 6.00 pm: CHECK-IN AND MIXER

6.00 – 7.30 pm: WELCOME DINNER

7:30 – 10:00 pm: Dr. Jasper A. Vrugt [Introduction to Machine Learning: Optional]

Wednesday, May 28, 2025

7.45 – 8.10 am: BREAKFAST

8.10 – 8.15 am: Dr. Jasper A. Vrugt / Dr. Efi Foufoula Georgiou [Opening Remarks]

8.15 – 9.15 am: Dr. Tapio Schneider, California Institute of Technology [Keynote Talk]

9.15 – 9.30 am: GROUP PHOTO

9.30 – 11.45 am: PRESENTATION BLOCK 1 [Chairs: Dr. Jasper Vrugt, Dr. Efi Foufoula-Georgiou]

9.30 – 9.45 am: **Dr. Dapeng Feng**, Stanford University: *Determining the relative influence of water potential, biomass, and temperature on vegetation optical depth with Explainable AI* [dpfeng@stanford.edu] **1**

9.45 – 10.00 am: **Nishchal Sigdel**, PhD Student, Virginia Tech: *Suspended Sediment Prediction with Long-Short Term Memory Models* [nishchalnsigdel@vt.edu] **1**

10.00 – 10.15 am: Yifu Gao, PhD Student, UC Irvine: *Soil Moisture Cloud Precipitation Feedbacks From Satellite Data Mining and Machine Learning* [yifug1@uci.edu] **1**

10.15 – 10.30 am: Dr. Chaopeng Shen, Pennsylvania State University: *State-of-the-Art AI & Physics-Informed ML in Hydrology and Beyond: Insights and Synergies* [shen.chaopeng@gmail.com] **1**

10.30 – 10.45 am: REFRESHMENTS

10.45 – 11.00 am: **Dr. Velimir (Monty) Vesselinov**, EnviTrace LLC: *GAIA: Cloud Framework for Geospatial Artificial Intelligence Analyses* [monty@envitrace.com] **1**

11.00 – 11.15 am: **Chuqiang Chen**, PhD Student, Virginia Tech: *Predicting runoff and baseflow hydrograph components at a continental scale with deep learning* [chuqiang@vt.edu] **1**

11.15 – 11.30 am: **Hesham Morgan**, PhD Student, Chapman University: *Integrated Geospatial Modeling and Machine Learning Approaches for Groundwater Potential Mapping in Hyper-Arid Regions* [hmorgan@chapman.edu] **1**

11.30 – 11.45 am: **Zihang Ding**, PhD Student, University of Minnesota, Duluth, *Optimizing Sensor Placement for Urban Flood Reconstruction: A Data-Driven Sparse Sensing Approach* [ding0430@umn.edu] **1**

11.45 – 12.00 pm: DISCUSSION BLOCK 1

12.00 – 1.15 pm: LUNCH

1.15 – 2.15 pm: Dr. Jim Randerson, University of California Irvine [Plenary Talk]

2.15 – 3.00 pm: PRESENTATION BLOCK 2 [Chair: Dr. Efi Foufoula-Georgiou]

2.15 – 2.30 pm: **Aramide Moronfoye**, PhD Student, University of Illinois at Urbana-Champaign: *Explaining Graph Neural Network Forecasts of Spatiotemporal Root Zone Soil Moisture using a Generative, Counterfactual Approach* [aramide2@illinois.edu] **1**

2.30 – 2.45 pm: **Dr. Somnath Bar**, UC Irvine: *Building Flammability and Atmospheric Dryness as Dominant Predictors of Wildfire Building Damage: A Data Driven Approach Across California's WUI* [somnatb@uci.edu] **1**

2.45 – 3.00 pm: **Shu Li**, PhD Student, UC Irvine: *Enhancing Fire Behavior Modeling with Live Fuel Moisture Analysis under Climate Change* [shul15@uci.edu] **1**

3.00 – 4.00 pm: PRESENTATION BLOCK 3 [Chair: Dr. Xingyuan Chen]

3.00 – 3.15 pm: **Dr. Timothy Hodson**, U.S. Geological Survey: *Multivariate model evaluation and multi-model inference with normalizing flows* [thodson@usgs.gov] **2**

3.15 – 3.30 pm: **Dr. Sina Jahangir**, McGill University & Cornell University: *Knowledge distillation for hydrological prediction* [mj649@cornell.edu] **2**

3.30 – 3.45 pm: **Xiaofeng Liu**, Michigan Institute for Data and AI in Society, University of Michigan: *Identifying Trustworthiness Challenges in Deep Learning Models for Continental-Scale Water Quality Prediction* [xiaofeng.liu0202@gmail.com] **3**

3.45 – 4.00 pm: **Dr. Jasper A. Vrugt**, UC Irvine**:** *Bayesian Uncertainty Analysis Under Misspecification* [jasper@uci.edu] **3**

4.00 – 4.15 pm: REFRESHMENTS/DISCUSSION BLOCKS 2 AND 3

4.15 – 6.00 pm: POSTER SESSION [Lakeview]

1. Ryoko Araki, PhD Student, San Diego State University & UC Santa Barbara: *Random Forest Model Predictions of Hydrologic Signatures for Enhanced Process Understanding* [raraki8159@sdsu.edu] **1**

2. Tarun Agrawal, PhD Student, University of Illinois at Urbana Champaign: *Integrating flow and flux dynamics in an adaptive LSTM model for stream chemistry predictions* [taruna2@illinois.edu] **1**

3. Mohammad Ali Farmani, PhD Student, University of Arizona, Tucson: *Accelerating Noah-MP Calibration: A Physically Informed Surrogate Model for Machine Learning Integration* [farmani@arizona.edu] **1**

4. Diver Eduardo Marin Palacio, PhD Student, Virginia Tech: *A continental-scale long short-term memory model for predicting freshwater salinization* [diverm@vt.edu] **1**

5. Dr. Yeonjoo Kim, Yonsei University: *Assessing Reservoir Construction Strategies under Changing Climate Using Differentiable Modeling* [yeonjoo.kim@yonsei.ac.kr] **1**

6. Ann Scheliga, PhD Student, UC, Berkeley: *CYGNSS surface water maps to improve LSTM streamflow prediction downstream* [ann_scheliga@berkeley.edu] **1**

7. Dr. Mashrekur Rahman, USDA-ARS & UC Davis: *Drivers of Soil Moisture Dynamics over Continental United States* [mashman@ucdavis.edu] **1**

8. Dr. Runze Li, UC Irvine: *Added value of environmental variables for satellite precipitation retrieval: A temporal coevolution perspective and a machine learning integration assessment* [runzl11@uci.edu] **1**

9. Dr. Amit Kumar, University of Minnesota Duluth: *Enhancing Total Phosphorus Prediction in Streams Using Hybrid Machine Learning* [kuma0341@d.umn.edu] **4**

10. Robert Keppler, PhD student, Geological Survey of Denmark and Greenland (GEUS): *Learning shallow water equations with physics-informed Deep Operator Network (DeepONet)* [rokep@geus.dk] **4**

11. Tao Huang, PhD Student, Boise State University: *Applying Machine Learning Models to Assess Vegetation Cover and Stream Temperature in Idaho* [taohuang@u.boisestate.edu] **4**

12. Surendra Maharjan, PhD Student, Chapman University: *Physics-Informed Machine Learning for Assessing the Impact of Phase of Precipitation in Snow-Fed Catchments* [maharjan@chapman.edu] **4**

13. Dr. Yuan Yang, Center for Western Weather and Water Extremes, Scripps Institution of Oceanography, UC San Diego: *Global Daily Discharge Estimation Based on Grid Long Short-Term Memory (LSTM) Model and River Routing* [yuanyangthu@gmail.com] **4**

14. Hossein Yousefi Sohi, PhD Student, University of Arizona, Tucson: *Stochastic-Deterministic Fusion: A Generative Downscaling Framework for High-Resolution Atmospheric Forecasting and Hydrologic Applications* [hyousefi@arizona.edu] **4**

15. Omid Zandi, PhD Student, University of Arizona, Tucson: *UA-HIPA V1: University of Arizona High Latitude Infrared-based Precipitation retrieval algorithm using AVHRR sensor Version 1* [zandiomid1992@gmail.com] **4**

16. Dr. Mu Xiao, CW3E, UC San Diego: *LSTM-based Post-Processing Improves Streamflow Prediction in the Sierra Nevada* [mxiao@ucsd.edu] **4**

17. Shahryar Fazli, PhD Student, Chapman University: *Machine Learning-Driven Water Use Efficiency Forecasting Across Western U.S. Agricultural Regions with CMIP6 Integration* [sfazli@chapman.edu] **4**

6.00 – 7.30 pm: DINNER

8.00 – 9.00 pm: HydroML-BINGO WITH PRIZES!

Thursday, May 29, 2025

7.45 – 8.30 am: BREAKFAST

8.30 – 9.30 am: Dr. Praveen Kumar, University of Illinois, Urbana-Champaign [Plenary Talk]

9.30 – 11.45 am: PRESENTATION BLOCK 4 [Chairs: Ajinkya Desai, Dr. Tirtha Banerjee]

9.30 – 9.45 am: **Dr. Xingyuan Chen**, Pacific Northwest National Laboratory: *Shaping the digital future: reimagining the scientific method with AI* [Xingyuan.Chen@pnnl.gov] **4**

9.45 – 10.00 am: **Tim Johnsen**, UC Irvine: *Evapotranspiration Prediction, Uncertainty Quantification, Denoising, and Gap Filling with Considerations in Sensor Placement, Fusion, Spatial and Temporal Correlations* **[**tjohnsen@uci.edu] **4**

10.00 – 10.15 am: **Dr. Haokai Zhao**, Massachusetts Institute of Technology: *Integration of Advanced Wireless Sensing with Physics-Based Machine Learning for Real-Time Soil Moisture Estimation and Forecasting* [hkzhao7@mit.edu] **4**

10.15 – 10.30 am: **Dr. Shubhendu Kumar Singh**, Los Alamos National Laboratory: *Deep Reservoir: Deep Reinforcement Learning (DRL) for Optimal Control of Water Reservoirs* [shubhsingh@lanl.gov] **4**

10.30 – 10.45 am: REFRESHMENTS

10.45 – 11.00 am: **Anna Valcarcel**, PhD Student, University of California, Berkeley: *Streamflow Estimation for Tropical Ungauged Basins Using CYGNSS-Based Surface Water Maps and Neural Networks* [arvalcarcel@berkeley.edu] **4**

11.00 – 11.15 am: **Dr. Sushree Swagatika Swain**, Scripps Institution of Oceanography, University of California San Diego: *Data-Driven Climate Forecasting: Machine Learning and Ensemble Models for Regional Climate* [sushreeswain1991@gmail.com] **4**

11.15 – 11.30 am: **Dr. Goutam Konapala**, Pacific Northwest National Laboratory: *Uncertainty-aware deep learning-based hydrologic modeling at the Continental United States scale* [goutam.konapala@pnnl.gov] **4**

11.30 – 11.45 am: Dr. Yuan-Heng Wang [online], Earth and Environmental Science Area, Lawrence Berkeley National Laboratory & Department of Hydrology and Atmospheric Science, University of Arizona: *Developing Catchment-Scale Rainfall-Runoff Modeling Using the Mass-Conserving-Perceptron-Based Neural Networks* [yhwang0730@gmail.com] **4**

11.45 – 12.00 pm: QUESTIONS/DISCUSSION BLOCK 4

12.00 – 1.15 pm: LUNCH

1.15 – 2.00 pm: DISCUSSION & CONCLUDING REMARKS

Adjourn